

NIRMA UNIVERSITY
Institute of Technology
MCA Computer Science and Engineering
Practical List
Semester-I

6CS102CC25	Data Structures	[3-0-2-4]
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Sr. No.	Practical Title		Hours	CLO
1.	(i)	An education institute has to maintain a list of courses it offers. There is a need to access the list of courses frequently. The existing courses are rarely removed, while new courses are often added. Considering this information, use an appropriate data structure to implement a system that allows for adding new course, removing existing course from the system, and listing courses.	02	1, 4
	(ii)	Implement the following programs on LeetCode based on the concepts of array data structure (i.e, Single/Multidimensional, matrix operations, etc). 1. https://leetcode.com/problems/two-sum/ 2. https://leetcode.com/problems/search-a-2d-matrix/		
2.	(i)	Write a program to determine balanced parentheses in a given algebraic expression, considering that only one type of parenthesis is used in an algebraic expression.	02	1, 2, 4
	(ii)	Implement the following programs related to the stack concept and its applications. 1. https://leetcode.com/problems/valid-parentheses/ 2. https://leetcode.com/problems/longest-valid-parentheses/		
3.	(i)	Write a program to evaluate a given infix expression.	04	1, 2, 4
	(ii)	Implement the following programs related to the stack concept and its applications.		

		https://leetcode.com/problems/remove-all-adjacent-duplicates-in-string/description/ https://leetcode.com/problems/validate-stack-sequences/description/		
	(iii)*	Write a program to implement the LIFO scheduling policy of CPU.	04	1, 2, 4
4.	(i)	Write a program to implement a call management system for incoming calls at a call center. The first call received is the first to be connected. Use an appropriate data structure to implement the system.	04	1, 2, 4
	(ii)	Implement the following programs related to the Queue data structure. https://leetcode.com/problems/design-circular-queue/ https://leetcode.com/problems/first-unique-character-in-a-string/description/		
	(iii)*	Write a program to perform the following operations on a Double-ended Queue. Implement a Double-ended Queue using static memory allocation. - Insert at Left - Insert at Right - Delete from Left - Delete from Right - Display	02	1, 2, 4
5.	(i)	Write a program to simulate heap memory management operations. Maintain two lists: free memory list and used memory list using dynamic memory management functions. Implement heap memory management operations: (a) allocate memory to the program (b) deallocate memory from the program (c) display a list of free nodes and (d) display a list of used nodes.	04	1, 2, 4
	(ii)	Implement the following programs based on the concepts linked- list. https://leetcode.com/problems/reverse-linked-list/ https://leetcode.com/problems/merge-two-sorted-lists/description/		

6.	(i)	Implement a music playlist application using the appropriate data structure. There is no estimation for the number of music files to be managed by this application. Users are expected to frequently add new music files and/or remove existing ones. Support functionalities for (a) playing the next file, (b) playing the previous file, (c) playing the first file and (d) playing the last file, and (e) playing the specific file.	02	1, 2, 4
	(ii)	Implement the following programs based on the concepts linked list. 1. https://leetcode.com/problems/design-linked-list/description/ 2. https://leetcode.com/problems/design-browser-history/description/		
	(iii)*	Write a program to display the string in reverse. Store each character of the string as a node in the linked list.	01	1, 2, 4
	(iv)*	Write a program to implement a Circular Queue using dynamic memory location.	01	1, 2, 4
7.	(i)	Implement <i>word to definition (dictionary/thesaurus)</i> using Binary Search Tree, which provides the following operations: (a) add a new entry in the dictionary, (b) remove the entry from the dictionary, (c) search word from dictionary, (d) list all entries in ascending order of word and (e) list all entries in descending order of word.	04	1, 2, 3, 4
	(ii)	Implement the following programs based on the Binary Tree data structure. 1. https://leetcode.com/problems/balance-a-binary-search-tree/ 2. https://leetcode.com/problems/minimum-absolute-difference-in-bst/		

8.	(i)	Create a simple graph to represent device network connectivity. Represent devices as nodes and network cables as edges. Devices and their connectivity are given below: • Device A is connected to Device B and Device C. • Device B is connected to Device C and Device D. • Device C is connected to Device D. • Device D is connected to Device E. • Device E is connected to Device A. Create an adjacency matrix for the graph. Indicate if the graph is directed or undirected.	02	1, 2, 4
	(ii)	Implement the following programs based on graphs. 1. https://leetcode.com/problems/find-center-of-star-graph/ 2. https://leetcode.com/problems/find-if-path-exists-in-graph/		
9.	(i)	Implement a Binary search technique, which takes a list of unique roll numbers that are sorted in descending order and a value to search for, and returns either the index of the value or -1 if the value isn't in the list.	02	1, 3
	(ii)	Implement the following programs. 1. https://leetcode.com/problems/search-in-rotated-sorted-array/ 2. https://leetcode.com/problems/find-minimum-in-rotated-sorted-array-ii/		
10.	(i)	Write a program to implement bubble sort and selection sort for sorting a given set of integers in ascending order and calculate time complexity.	04	1, 3
		Implement following programs. 1. https://leetcode.com/problems/merge-sorted-array/ 2. https://leetcode.com/problems/remove-duplicates-from-sorted-array/		
	(ii)*	Write programs for applying following operations on one-dimension array of integers: (1) Merge sort (2) Heap Sort	02	1, 3

Note: * indicates optional practical for practice